Lecture 20

Inversion and determinant definitions

Uses.

Summary 13: The cross-product length gives parallelogram area, and its oriented normal fixes the turn direction. The scalar triple product is signed volume; its absolute value is the parallelepiped's volume.
Summary 17: Row operations preserve solutions, and elimination terminates. A free-variable assignment determines a unique solution of a consistent reduced system.
Summary 18: The null space contains all homogeneous solutions. Its special free-variable columns give a basis. Rank is the cited intrinsic pivot count.
Summary 19: Compatible products represent composition. Multiplication is associative. Elementary matrices are invertible and act by row operations. An inverse is unique and satisfies both product identities.

Objectives. *Elimination decides whether a square matrix has an inverse and, when it does, constructs that inverse. The same calculation solves a system for every right side. We then define the determinant by first-column expansion and identify its real geometric meaning in dimensions two and three. The relation between determinants and invertibility is proved in Lecture 21.*

By the end of this lecture you will be able to:

- Prove the equivalent elimination and homogeneous-system criteria for a square matrix to be invertible.
- Construct an inverse by row operations on the full augmented array.
- Compute a determinant from its recursive first-column definition.
- Interpret two- and three-dimensional determinants as signed area and volume.

20.1 Inverse products and the elimination criteria

Let $K = \mathbb{R}$ or $K = \mathbb{C}$ and let $A \in K^{n\times n}$, where $n \geq 1$. By Lecture 19, an inverse satisfies $AA^{-1} = A^{-1}A = I_n$. Both identities are part of the definition. A single identity will not be used as a substitute for the definition in this lecture. Sheet 20 proves the additional square-matrix result that one identity implies the other.

First, if U and V are invertible matrices of the same square size, then their product is invertible and its inverse is

$$(UV)^{-1} = V^{-1}U^{-1}. \tag{20.1}$$

To prove this, use associativity from Lecture 19 in both orders:

$$(UV)(V^{-1}U^{-1}) = U(VV^{-1})U^{-1} = I_n, \qquad (V^{-1}U^{-1})(UV) = V^{-1}(U^{-1}U)V = I_n.$$

Thus the proposed matrix satisfies the two defining identities. Repeated application shows that a finite product of invertible matrices is invertible. Its inverse has the reversed order of the individual inverses. The product of no factors is defined to be I_n and has inverse I_n.

Choose any sequence of elementary row operations that reduces A to a reduced row-echelon matrix R. If their elementary matrices are $E_1, \dots, E_s$, put

$$E = E_s \cdots E_1, \qquad R = EA.$$

Every factor is invertible by the reverse-operation argument of Lecture 19. Therefore E is invertible by Eq. (20.1). For a sequence with no operations, take $E = I_n$.

Theorem 20.1 (Square-matrix inversion criteria). *For $A \in K^{n\times n}$, the following conditions are equivalent:*

1. *The matrix A is invertible.*
2. *The homogeneous system $Ax = 0_n$ has only the solution $x = 0_n$.*
3. *A reduction of A has n coefficient pivots.*
4. *The reduced row-echelon matrix obtained from A is I_n.*

When these conditions hold, A is a product of elementary matrices.

Proof. If A is invertible and $Ax = 0_n$, multiply by A^{-1} to get $x = A^{-1}Ax = A^{-1}0_n = 0_n$. Thus invertibility implies the homogeneous condition.

Suppose that homogeneous condition holds. If a chosen reduction had a free column, assign 1 in that free coordinate and 0 in the other free coordinates. The consistency theorem of Lecture 17 would give a nonzero homogeneous solution, a contradiction. Hence all n columns have pivots. Conversely, with n coefficient pivots there are no free columns, and the homogeneous equations force the zero solution by the same theorem.

In an $n \times n$ matrix, n pivots occupy every row and column. In reduced form each pivot is 1 and its column has no other nonzero entry. Their left-to-right order therefore gives exactly $R = I_n$. The identity matrix also has n pivots, proving the converse connection.

Finally, if $EA = I_n$ for the invertible product E defined above, left multiplication by E^{-1} gives $A = E^{-1}$. The two identities $AE = EA = I_n$ now follow, so A is invertible with inverse E. Moreover, $A = E_1^{-1} \cdots E_s^{-1}$ is a product of elementary matrices because each inverse is the elementary matrix of the reverse operation. For $A = I_n$ the empty product gives the stated conclusion. These implications prove all equivalences. ∎

This proof uses an actual reduction. It does not use a determinant. Writing the pivot condition as $\operatorname{rank} A = n$ uses the rank convention of Lecture 18, whose route-independent meaning inherits the cited input of Lecture 17. The proof above gives the same answer for every actual reduction.

An invertible matrix also makes $Ax = b$ uniquely solvable for every $b \in K^n$. The solution is

$$x = A^{-1}b.$$

It satisfies $Ax = AA^{-1}b = b$. Every solution must have that value after multiplying its equation by A^{-1}. Conversely, unique solvability for every right side includes the homogeneous case $b = 0_n$, so Theorem 20.1 proves invertibility. Thus invertibility decides all right sides at once.

20.2 Gauss–Jordan inversion

To construct the inverse, attach I_n to the square coefficient matrix:

$$[A \mid I_n].$$

Reduce the left block by elementary row operations, applying every operation to the entire augmented row. After a sequence with product E, the blocks are $[EA \mid E]$, since the initial right block is I_n. If the left block reaches I_n, Theorem 20.1 proves $E = A^{-1}$. The final array is therefore $[I_n \mid A^{-1}]$. If the reduced left block is not I_n, the same theorem proves that no inverse exists. The procedure stops with that decision; pivots in the right block do not supply missing coefficient pivots.

For a full example, invert the matrix

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{pmatrix}.$$

Start from

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 2 & 0 & 1 & 0 & 0 & 1 \end{array}\right).$$

First replace R_3 by $R_3 - 2R_1$. Then replace R_1 by $R_1 - 2R_2$ and R_3 by $R_3 + 4R_2$. The whole array becomes

$$\left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & -2 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 5 & -2 & 4 & 1 \end{array}\right).$$

Divide the third row by 5. Add twice the new third row to the first, and subtract it from the second. The result is

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/5 & -2/5 & 2/5 \\ 0 & 1 & 0 & 2/5 & 1/5 & -1/5 \\ 0 & 0 & 1 & -2/5 & 4/5 & 1/5 \end{array}\right).$$

All three coefficient pivots exist. The theorem identifies the right block as

$$A^{-1} = \frac{1}{5}\begin{pmatrix} 1 & -2 & 2 \\ 2 & 1 & -1 \\ -2 & 4 & 1 \end{pmatrix}.$$

The row-operation proof verifies both inverse identities through $A = E^{-1}$; it does not assume that the observed left identity alone is sufficient. For the right side $b = (3, 1, 4)$, the solution is

$$x = A^{-1}b = \frac{1}{5}\begin{pmatrix} 3 - 2 + 8 \\ 6 + 1 - 4 \\ -6 + 4 + 4 \end{pmatrix} = \begin{pmatrix} 9/5 \\ 3/5 \\ 2/5 \end{pmatrix}.$$

Its original left sides are $9/5 + 6/5 = 3$, $3/5 + 2/5 = 1$ and $18/5 + 2/5 = 4$, as required.

Example 20.2 (Elimination over the complex numbers)**.** Invert the complex matrix

$$C = \begin{pmatrix} 1+\mathrm{i} & 2 \\ 1 & -\mathrm{i} \end{pmatrix}.$$

Exchange the two rows of $[C \mid I_2]$ and subtract $1+\mathrm{i}$ times the new first row from the second. The resulting array is

$$\left(\begin{array}{cc|cc} 1 & -\mathrm{i} & 0 & 1 \\ 0 & 1+\mathrm{i} & 1 & -1-\mathrm{i} \end{array}\right).$$

The second pivot is nonzero, with reciprocal $(1-\mathrm{i})/2$ by Lecture 15. Divide the second row by $1+\mathrm{i}$ and add i times that row to the first. The left block is now I_2 and the right block gives

$$C^{-1} = \begin{pmatrix} (1+\mathrm{i})/2 & 1-\mathrm{i} \\ (1-\mathrm{i})/2 & -1 \end{pmatrix}.$$

The same proved construction applies over $\mathbb{C}$ because every row operation and its reverse uses the complex arithmetic laws. No ordering of the pivots as positive or negative numbers is required.

A failure of invertibility

Now take the square matrix

$$B = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 3 & 1 \end{pmatrix}.$$

Subtract its first and second rows from its third row, and then clear the entry 2 above the second pivot. Its reduced coefficient matrix is

$$R = \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

The free third coordinate gives the nonzero homogeneous solution $h = (-2, 1, -1)$: the original left sides are $-2+2 = 0$, $1-1 = 0$ and $-2+3-1 = 0$. By Theorem 20.1, the matrix B has no inverse. For a right side (b_1, b_2, b_3), its third left side is always the sum of the first two. Thus consistency requires $b_3 = b_1 + b_2$. For instance, the right side $(0, 0, 1)$ is inconsistent, while the zero right side has infinitely many solutions th, $t \in K$. Both behaviours agree with failure of invertibility.

20.3 The recursive determinant

For a square matrix $A \in K^{n\times n}$, its *determinant* is a scalar written $\det A$. If $n = 1$, define $\det(a_{11}) = a_{11}$. For $n \geq 2$, let A_{i1} be the $(n-1)\times(n-1)$ matrix obtained by deleting row i and column 1. Define the determinant recursively by expansion along the first column:

$$\det A = \sum_{i=1}^{n} (-1)^{i+1} a_{i1} \det A_{i1}. \tag{20.2}$$

The smaller determinants have already been defined by the recursion. The procedure ends at size 1, so it defines exactly one scalar for every positive square size. For later formulas

involving a deleted row and column of a 1×1 matrix, we also set the determinant of the empty 0×0 matrix to 1. With that convention the same expansion works at size 1 as well. No determinant is assigned here to a nonsquare matrix.

At size 2 the definition gives

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - cb.$$

At size 3, applying the same definition once and then the size-2 formula gives

$$\det \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & k \end{pmatrix} = a(ek - fh) - d(bk - ch) + g(bf - ce).$$

The signs along the first column are $+, -, +$. Every smaller determinant retains the original order of its remaining rows and columns. No row swap is implicit in deleting a row and column.

For the earlier invertible example, the definition gives

$$\det A = 1(1 \cdot 1 - 1 \cdot 0) - 0(2 \cdot 1 - 0 \cdot 0) + 2(2 \cdot 1 - 0 \cdot 1) = 5.$$

The number 5 also occurred as the final coefficient pivot in this particular elimination. We have not used that observation to prove a general determinant rule. Lecture 21 proves the row-operation rules and the invertibility criterion from this definition.

A matrix is *upper triangular* if all entries strictly below the diagonal are zero. For such a matrix the first-column expansion has only its first term, and its remaining minor is upper triangular again. Induction, starting from size 1, therefore gives

$$\det A = a_{11}a_{22} \cdots a_{nn} \quad \text{for upper triangular } A.$$

In particular $\det I_n = 1$, including the stated empty convention. This calculation is a direct use of the definition, without row operations.

20.4 Signed area and volume

For real planar columns $u = (u_1, u_2)$ and $v = (v_1, v_2)$, extend them to $\widetilde{u} = (u_1, u_2, 0)$ and $\widetilde{v} = (v_1, v_2, 0)$ in $\mathbb{R}^3$. The cross-product formula of Lecture 13 gives

$$\widetilde{u} \times \widetilde{v} = (0, 0, u_1v_2 - u_2v_1).$$

Its length is the parallelogram's ordinary area by that lecture. With the positive normal in the third coordinate direction, define the *signed area* to be the third component $u_1v_2 - u_2v_1$. The size-2 determinant formula therefore gives signed area $\det[\, u \; v \,]$ and ordinary area $|\det[\, u \; v \,]|$. Zero or parallel columns give zero area.

Figure 20.1 shows a positive planar orientation.

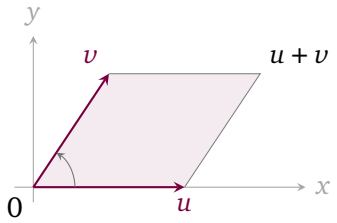


Figure 20.1: The ordered planar sides turn from u towards v counterclockwise through an angle between zero and π. Their signed area is positive; the absolute determinant gives area for either orientation.

For real columns $u, v, w \in \mathbb{R}^3$, Lecture 13 gives signed volume $u \cdot (v \times w)$. Expanding the determinant with these columns gives

$$\begin{aligned} \det[\, u\ v\ w\,] &= u_1(v_2w_3 - v_3w_2) - u_2(v_1w_3 - v_3w_1) + u_3(v_1w_2 - v_2w_1) \\ &= u \cdot (v \times w). \end{aligned}$$

The last equality is the previously derived component formula for the scalar triple product. Thus the determinant is signed volume, and its absolute value is the ordinary volume, including zero for coplanar columns.

In the running example, the three columns are $u = (1, 0, 2)$, $v = (2, 1, 0)$ and $w = (0, 1, 1)$. Their cross product and signed volume are

$$v \times w = (1, -2, 2), \qquad u \cdot (v \times w) = 1 + 0 + 4 = 5.$$

The parallelepiped therefore has volume 5 in the corresponding coordinate volume units. This agrees with the computed determinant. The area and volume interpretations concern real columns. The recursive determinant itself remains defined over $\mathbb{C}$ by the same finite arithmetic, without assigning a planar-area meaning to its complex value.

Summary 20.

- Equation (20.1): invertible products have inverse factors in reverse order. Both defining inverse identities are verified.
- Theorem 20.1: for square A, invertibility, a zero null space, n pivots and reduced form I_n are equivalent. Such a matrix is a product of elementary matrices and solves every right side uniquely.
- Reducing the left block of $[A \mid I_n]$ to I_n produces $[I_n \mid A^{-1}]$. A missing coefficient pivot means that no inverse exists.
- Equation (20.2): the determinant is defined by first-column expansion, with the stated size-1 and empty conventions. Upper triangular matrices have the product of their diagonal entries as determinant.
- For real columns, $\det[\,u\ v\,]$ is signed area and $\det[\,u\ v\ w\,] = u \cdot (v \times w)$ is signed volume. Their absolute values give ordinary area and volume.

Sheet 20 practices inversion and determinants and proves the promised one-sided inverse result. Lecture 21 proves that a nonzero determinant is equivalent to invertibility.

References

D. C. Lay, S. R. Lay and J. J. McDonald, *Linear Algebra and Its Applications*, 5th ed., Pearson, 2016, §§2.2–2.3 and 3.1. S. Lang, *Introduction to Linear Algebra*, 2nd ed., Springer, 1986, Chapter VII, §1.